A fourth configuration with stationary orthocenter

MathDB 365004 (duplicate formulation 372155)

Principia Math In Progress

21 September 2026

Task 1: a full counterexample

The classification conjecture of Garcia, Helman and Reznik [1, Conjecture 1, p.6] asserts that a Poncelet triangle family can have a stationary orthocenter only in the focal-X_4, MacBeath, or Dual configurations. The following counterexample has nested, noncircular ellipses and consists entirely of acute triangles.

Theorem 1. *The two ellipses*

$$E: \ \frac{x^2}{4} + y^2 = 1, \qquad K: \ \frac{(x - 31/20)^2}{4/25} + \frac{y^2}{63/1600} = 1$$

support a full Poncelet family of triangles. Every triangle has orthocenter $H = (31/16, 0)$*. This conic pair is none of the three configurations in the conjecture.*

Credit and prior work. The general orthocenter-locus semiaxis formula is already [2, Theorem 3]; the example can be obtained by specializing that formula. We explicitly credit it and give a direct algebraic verification below. The campaign assigns **zero new credit** pending inspection of the identified journal final of the original source [3]. No new general locus formula is claimed.

1. Nesting and the full family

Apply $(x, y) = (2X, Y)$ to the unit circle. The inverse image of K is

$$K_0: \quad (X, Y) = \left(\frac{31}{40} + \frac{1}{5}\cos t, \frac{3\sqrt{7}}{40}\sin t \right).$$

With $v = \cos t$ its squared distance to the origin is

$$X^2 + Y^2 = \frac{v^2 + 496v + 1024}{1600} \leq \frac{1521}{1600} < 1.$$

The polynomial is increasing on $[-1, 1]$. Thus the caustic is strictly nested, and both its semiaxes are positive.

Put $f = 4/5$, $g = 3/4$, $s = f + g = 31/20$ and $p = fg = 3/5$. For every $|\lambda| = 1$, let a, b, c be the roots of

$$P_\lambda(z) = z^3 - (s + p\lambda)z^2 + (p + s\lambda)z - \lambda. \tag{1}$$

The equation $P_\lambda(z) = 0$ is equivalent on the unit circle to

$$B(z) = \lambda, \qquad B(z) = \frac{z(z-f)(z-g)}{(1-fz)(1-gz)}.$$

This degree-three finite Blaschke product has strictly positive argument derivative $1+(1-f^2)/|e^{it}-f|^2 + (1-g^2)/|e^{it}-g|^2$. Its total argument change is 6π. Consequently every λ has exactly three distinct roots on the unit circle. The triangles with vertices $Z(a), Z(b), Z(c)$, where $Z(z) = (3z + z^{-1})/2$, are nondegenerate and inscribed in E.

2. Every side touches the same caustic

For a pair a, b write $q = a + b$, $r = ab$. Vieta's formulas from (1) imply

$$c(1 - pr) = s - q, \qquad r - p = c(sr - q).$$

Here $1 - pr \neq 0$, since $|pr| = 3/5$. Eliminating c gives

$$q^2 - \frac{31}{20}(1 + r)q + \frac{3}{5}(1 + r^2) + \frac{417}{400}r = 0. \tag{2}$$

The chord of the unit circle through a, b has equation $z + r\bar{z} = q$. Write $r = e^{2i\theta}$ and $q = 2de^{i\theta}$ with $d \in \mathbb{R}$. This is the real line $X\cos\theta + Y\sin\theta = d$. Its tangency condition to K_0 is

$$\left(d - \frac{31}{40}\cos\theta\right)^2 = \frac{1}{25}\cos^2\theta + \frac{63}{1600}\sin^2\theta.$$

Multiplication by $4r \cdot 1600$ converts this condition into

$$1600q^2 - 2480q(r + 1) + 897(r + 1)^2 + 63(r - 1)^2 = 0.$$

This is precisely 1600 times (2). Every side therefore touches K_0, and its affine image touches K.

The caustic is inside each triangle. At $\lambda = 1$, the roots are 1 and $23/40 \pm 3i\sqrt{119}/40$, giving the physical triangle

$$(2, 0), \quad \left(\frac{23}{20}, \frac{3\sqrt{119}}{40}\right), \quad \left(\frac{23}{20}, -\frac{3\sqrt{119}}{40}\right).$$

Its interior contains the caustic center $(31/20, 0)$. As λ varies, roots remain distinct, and a tangent line cannot cross the center of a nondegenerate ellipse. Continuity preserves the interior side for all three supporting lines. Thus all these triangles circumscribe K. They form a full Poncelet family: every point of E occurs as a vertex, by choosing $\lambda = B(z)$ for its preimage z.

3. All altitudes pass through the fixed point

Put $A = 3/2$, $D = 1/2$, $h = 31/16$, $L = abc$, $s_1 = a + b + c$ and $s_2 = ab + bc + ca$. For unit a, b, c, $Z(a) = Aa + D/a$ and $\overline{Z(a)} = A/a + Da$. Direct expansion gives

$$\begin{aligned}&(h - Z(a))(\overline{Z(b)} - \overline{Z(c)}) + (h - \overline{Z(a)})(Z(b) - Z(c))\\ &\quad = \frac{b-c}{L}\left[(DL - Aa)h + (AL - Da)h - 2ADLa + (A^2 + D^2)s_1 a - (A^2 + D^2)s_2 + 2AD\right].\end{aligned}$$

Substitute $L = \lambda$, $s_1 = s + p\lambda$, $s_2 = p + s\lambda$. The bracket is identically zero: its a, λa, λ, and constant coefficients all vanish. The left side is twice the real Euclidean dot product of $H - Z(a)$ with $Z(b) - Z(c)$. Thus H lies on the altitude from $Z(a)$; cyclically it lies on all three. It is their orthocenter.

4. Exclusion of the three configurations

The definitions in [1, Sections 3–4] distinguish the three families by their actual Euclidean conic configurations:

Configuration	Reason our pair is different
MacBeath	The outer conic is a circle; E has semiaxes $2, 1$.
Focal-X_4	The inner conic is a circle; $4/25 \neq 63/1600$.
Dual	The two ellipses are concentric; their centers here differ.

These distinctions are preserved by similarities. In addition, H is not a focus of E, since $(31/16)^2 \neq 3$. Finally,

$$\frac{(31/16 - 31/20)^2}{4/25} = \frac{961}{1024} < 1.$$

So H lies strictly inside K and therefore inside every family triangle. Every triangle is acute. Neither degeneracy nor an obtuse exception is involved. This completes the counterexample.

Prior formula and audit status

For outer semiaxes a_0, b_0, the known formula [2, Theorem 3] has the semiaxis numerator

$$\sigma = \left| fg(a_0^2 + b_0^2) - (a_0^2 - b_0^2) \right|.$$

Our choice gives $\sigma = |(3/5)5 - 3| = 0$. Thus the example is consistent with that earlier theorem. Its application to the later proposed three-family classification is the subject of this note.

The full current 24-page original preprint, the complete 7-page classification preprint, and the complete 18-page general-locus preprint were read. The 32-page final journal article corresponding to the original preprint remains unread; the publisher provides a subscription preview. Source and priority eligibility therefore remain pending. MathDB 365004 and 372155 are the same problem, not two credits.

The accompanying verifier checks nesting, the exact cubic and tangency identities, and the universal altitude identity. It also checks 48 configurations independently in Euclidean coordinates at 90 decimal digits. These numerical checks supplement the proof. Review is by the authoring agent only.

Proposed follow-up Tasks 2–5

1. **Task 2.** Find a synthetic explanation for the stationary orthocenter of this pair.
2. **Task 3.** Classify all nested ellipse pairs with stationary orthocenter, up to similarities, using the credited general formula.
3. **Task 4.** Describe the exact region of attainable stationary orthocenters for a prescribed outer ellipse.
4. **Task 5.** Determine the polar-circle invariants of every family in that classification.

References

[1] R. A. Garcia, M. Helman, D. Reznik, *Four special Poncelet triangle families about the incircle*. arXiv:2512.17440v2 (2025).

[2] R. A. Garcia, M. Helman, D. Reznik, *Poncelet triangles: conic loci of the orthocenter and of the isogonal conjugate of a fixed point*. arXiv:2508.02368v3 (2025).

[3] R. A. Garcia, M. Helman, D. Reznik, *Harmonious loci of Poncelet triangles about the incircle and their degeneracies*. arXiv:2409.19464v7 (2025). Journal final under the title *Graceful loci*, Beitr. Algebra Geom. **67** (2026), 667–698, doi:10.1007/s13366-026-00825-1; final unread.